

RAEM RAHMAN

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EDUCATION

Harvey Mudd College <i>B.S. in Computer Science and Mathematics · GPA (major): 3.5</i>	Claremont, CA <i>Class of 2029 (Expected)</i>
Aitchison College <i>High School Diploma · GPA: 3.89</i>	Lahore, Pakistan <i>Class of 2025</i>

EXPERIENCE

AI / Machine Learning Engineer Intern <i>Mindigital</i>	Summer 2026 <i>Beijing, China (On-site)</i>
- Engineered a novel risk feature for a loan-default prediction model on a heavily biased dataset, lifting backtest PR-AUC from 0.78 to 0.83 (+0.05, a ~6% relative gain) using a gradient-boosted tree model (tested XGBoost, LightGBM, CatBoost), increasing default prediction accuracy and profitability. - Built a Docker-ized hybrid semantic + keyword search knowledge base application (Obsidian, pgvector) with agentic automation (Ollama, Hermes) for data logging and report generation, reducing manual reporting overhead and improving internal information retrieval — deployed on-premise for data-privacy compliance.	
Research Assistant <i>International Growth Centre (IGC)</i>	July 2024 - September 2024 <i>Lahore, Pakistan (Remote)</i>
- Built Python web-scraping, OCR and generative-AI pipelines to clean, gather, correlate, and analyze noisy large-scale power-sector data. - Contributed data analysis to the research paper “The Economic Costs of Lopsided Contracts: Evidence from Pakistan’s Power Sector,” quantifying financial losses from poorly conceived purchase power agreement (PPA) contracts.	
Artificial Intelligence & Software Intern <i>Office of State Senator Saud Anwar (D — Senate District 3)</i>	January 2024 - May 2024 <i>Hartford, CT (Remote)</i>
- Developed an automated pipeline generating daily batches of customized condolence letters to next of kin of constituents in Senate District 3. - Built obituary-search and letter-generation programs using Python, OpenAI APIs, and Scrapy for web scraping.	

TECHNICAL SKILLS

Languages & Tools: Python, SQL, Git
Machine Learning & AI: scikit-learn, PyTorch, NumPy, pandas, LangChain, RAG (semantic + keyword retrieval), OpenAI & Anthropic APIs, ReportLab, PyPDF
Data & Infrastructure: Docker, PostgreSQL / pgvector, Parquet, Ollama, Hermes, MySQL, Obsidian, BeautifulSoup, Scrapy
Spoken Languages: English (fluent), Hindi / Urdu (fluent), Chinese (HSK Level 3 Certified)

HONORS & AWARDS

Top 10%, Calculus — Stanford Math Tournament. Winner — Pakistan Mathematics Modeling Competition; selected for the Pakistan National Mathematics Modeling Team (2024). Top 40 nationally — invited to the Pakistan National Mathematics Team (IMO) Selection Camp, reserved for the top 40 ranked mathematics students in the country. 1st globally — Caribou Computer Science Coding Competition (2024). 1st in Pakistan — South East Asian Math Olympiad (SEAMO); invited to the global finals in Singapore (2024).
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